Research Update Summary

Optimization-Selected Uncertainty Quantification

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Current research question

The current formulation studies uncertainty quantification when a hyperparameter θ indexing an uncertainty set $\mathcal{C}(X; \mathcal{D}, \theta)$ is chosen by constrained optimization on the *same* calibration data $\mathcal{D}$ used to construct the set. Fixed-hyperparameter coverage need not survive this data-dependent choice. The project asks (i) when validity remains after data-dependent selection, (ii) how far actual miscoverage can deviate from the nominal level, and (iii) whether coverage can be restored by randomizing the *selection* map (stabilizing randomness ξ) while retaining the existing fixed-θ construction of $\mathcal{C}$.

Precision. The map $\mathcal{C}(\cdot\,; \mathcal{D}, \theta)$ for fixed, prespecified θ is unchanged. What changes is the procedure that produces θ (deterministic $\hat{\theta}(\mathcal{D})$ vs. randomized $\widetilde{\theta}(\mathcal{D}; \xi)$). Changing the selected hyperparameter generally changes the realized set.

Intellectual progression

Adaptive certified regret → stabilized-algorithm catalog → broad optimization-based selection stability framework (Formulation I; student-developed) → current optimization-selected UQ formulation (Formulation II). Full narrative: `Intellectual_Timeline.pdf`.

Main accomplishments

In order of intellectual development (subject to advisor feedback):

- **Mathematical Formulation I** — broad stability framework for post-hoc / optimization-based selection.
- **Stabilized-algorithm catalog** — structured extraction of stabilized selectors and recurring mechanisms.
- **Mathematical Formulation II** — Parts I–II agenda (`Formulation/writeup/Problem_Writeup.pdf`).
- **Discovery architecture** — supporting infrastructure for operator propositions.
- **Operator library and formal verification support** — selection-stability primitives with Lean certificates (supporting evidence; not the Part I–II chain).
- **Repository organization and advisor-facing documentation.**

Approximately **92 hours** reconstructed (June 1–July 29); detailed effort breakdown is in the intellectual timeline.

Current theorem agenda

- **Part I.** Replace-one stability of the optimization-selected hyperparameter $\hat{\theta}(\mathcal{D})$ under explicit assumptions.
- **Part II.** Validity and miscoverage after data-dependent selection; restore coverage by randomizing selection (no recalibration of $\mathcal{C}$). Part I informs diagnosis (i)–(ii); goal (iii) modifies the selection map.

Immediate next step

Complete the Part I theorem under explicit assumptions, obtain advisor feedback on formulation and assumptions, and only then extend.

What I would like to discuss

Concrete questions are in **`Questions_for_Advisors.pdf`** (formulation fit, novelty baselines, sequencing, collaboration). Please treat that one-pager as the discussion agenda.